Solution Approach & Design Explanation

Solution Overview

Our solution is a **Smart Telehealth System** that provides healthcare support in refugee or low-resource areas.
It ensures **patient identity, medical history tracking, and remote consultation**, even without internet.

Problem Approach

We solve the problem by:

- Creating a **portable health identity** for each patient
- Enabling **offline data storage and sync**
- Allowing **doctors to access records anytime**
- Using **AI support for basic diagnosis**

System Architecture

The system consists of:

- **User Module** (Patients & Doctors)
- **Data Storage Module** (Local + Cloud sync)
- **AI Module** (Health suggestions)
- **Communication Module** (Teleconsultation)

Working Process (Step-by-Step)

Step 1: Patient registers and gets a unique digital health ID
Step 2: Medical data is stored locally on device
Step 3: Doctor accesses patient data via app
Step 4: AI gives basic health suggestions
Step 5: When internet is available, data syncs to cloud

Design Components

Hardware (Optional)

- Mobile phone / tablet
- Wearable health sensors (optional)

Software

- Mobile App (Frontend)
- Database (Local + Cloud)
- AI Model for health prediction

Key Features

- Offline functionality
- Secure patient identity
- Easy doctor access
- AI-based assistance
- Data synchronization

Innovation / Uniqueness

- Works **without internet**
- Designed for **refugee camps & rural areas**

- Combines **AI + Telehealth + Identity system**

Future Scope

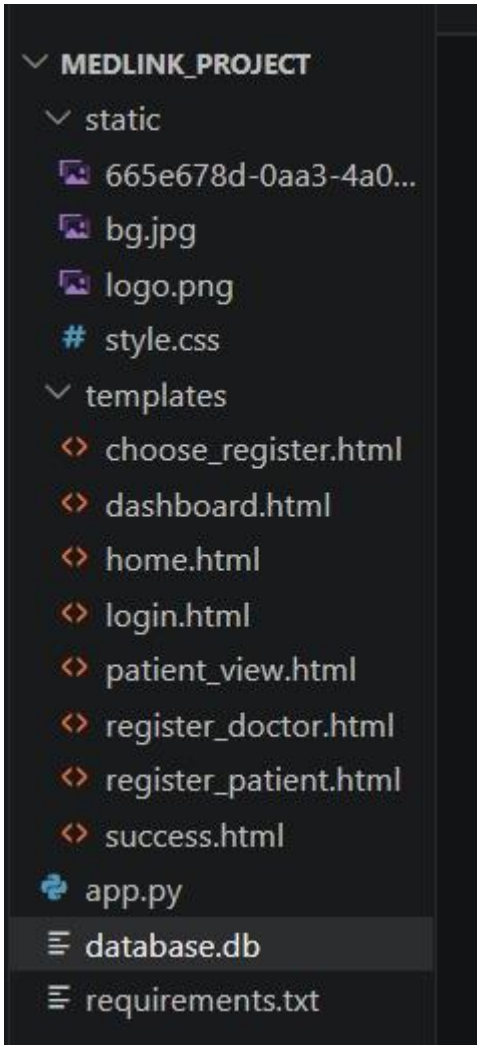
- Add wearable device integration
- Improve AI accuracy
- Expand to global healthcare systems
- Government & NGO collaboration

Arrow Flow

Flow Representation:

Register Patient → Store Data (Offline) → Doctor Access → AI Support → Sync to Cloud

Architecture Diagram



How the Solution Works

Step 1: User Registration / Identity Creation
A refugee is registered once using basic details and a unique digital health ID (QR/biometric/phone-based ID).

Step 2: Health Data Collection
Doctors or volunteers record patient symptoms, medical history, and treatment details during each visit.

Step 3: Local Data Storage (Offline Mode)
All health records are stored securely on a local device/server inside the camp without requiring internet.

Step 4: Medical Access Anytime
When a patient visits any clinic or NGO doctor, their health ID is used to instantly retrieve past records offline.

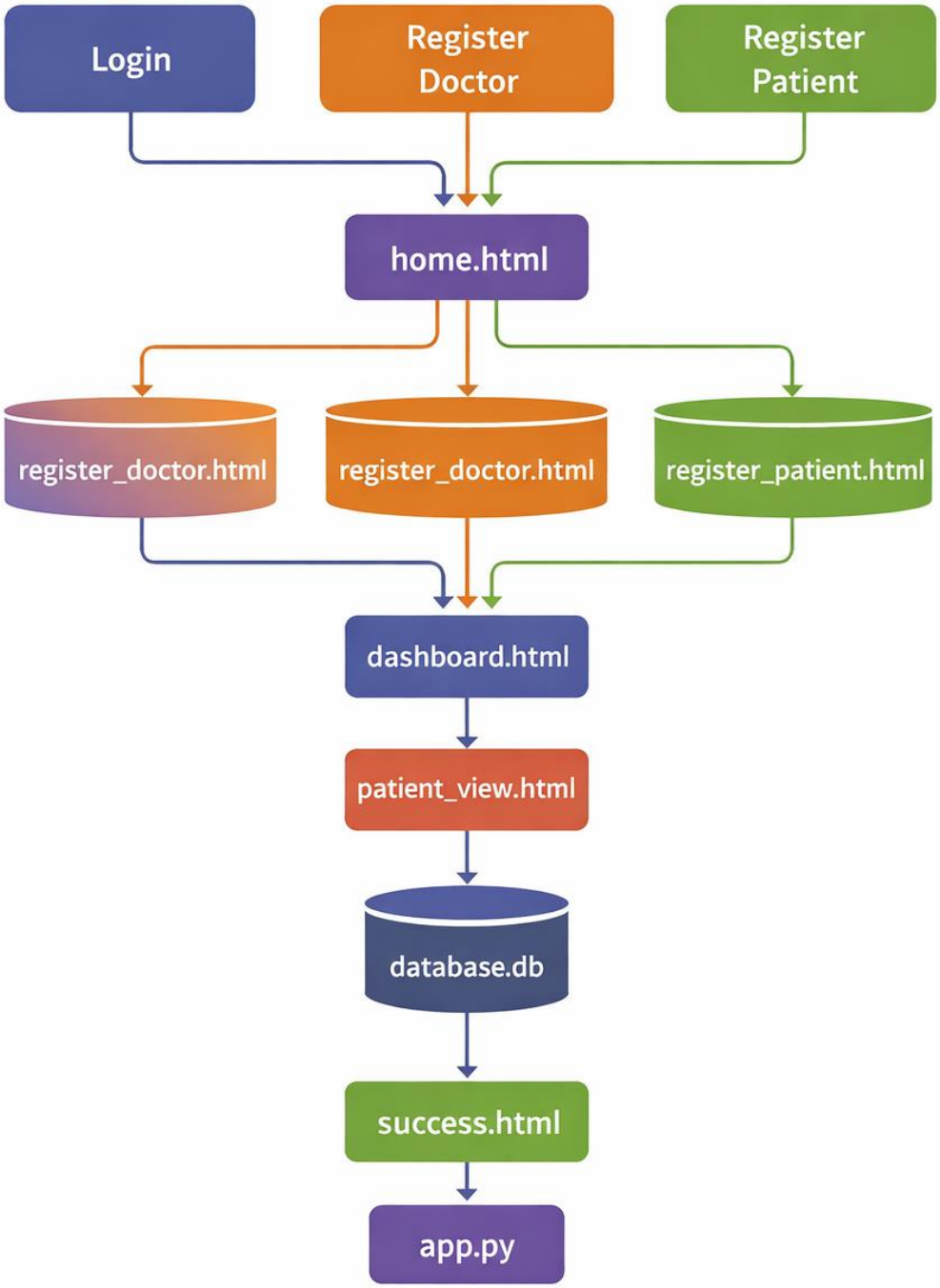
Step 5: Offline-to-Online Sync
Whenever internet is available, the system automatically syncs updated records to a central cloud database.

Step 6: Multi-Doctor Continuity
Different doctors (from different NGOs) can access the same updated patient history, ensuring continuous treatment.

Step 7: Emergency Access Feature
In emergencies, doctors can quickly view critical health information (allergies, conditions, medications) using the ID.

Step 8: Data Security & Backup
All data is encrypted and regularly backed up during sync to prevent loss or misuse.

Flow Chart Diagram


Login
Register Doctor
Register Patient
home.html
register_doctor.html
register_doctor.html
register_patient.html
dashboard.html
patient_view.html
database.db
success.html
app.py

Conclusion

The proposed Smart Telehealth System provides an efficient and reliable healthcare solution for refugee and low-resource communities. It ensures continuous medical support through offline data storage, secure patient identity, and AI-based health assistance. Even without internet connectivity, the system allows doctors to access patient records and provide timely treatment. When connectivity is available, data is synchronized to the cloud for better coordination. Overall, this system improves healthcare accessibility, reduces delays in treatment, and supports better decision-making in critical situations.